

TOI-4087b

R-2223

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_250702 · created 2026-08-14T23:58:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460858.716 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.039
Transit depth (Rp/Rs)^2	1.8 +/- 1.04 [%]
Orbital Inclination (inc)	88.91 +/- 2.06
Airmass coefficient 1 (a1)	0.931 +/- 0.022
Airmass coefficient 2 (a2)	0.04 +/- 0.11
Scatter in the residuals of the lightcurve fit is	2.43 %
Best Comparison Star	#3 - [393, 352]
Optimal Aperture	2.68
Optimal Annulus	10.2
Transit Duration (day)	0.056 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.134 $\pm$ 0.039 vs 0.10754068997360952 (deviation 0.68 σ)
Duration fit vs published	0.056 d vs 0.1242 d (deviation 5.68 σ)
Mid-time O-C	-15.1 min
Depth SNR	3.4
Residual scatter	2.43 %

Light curve

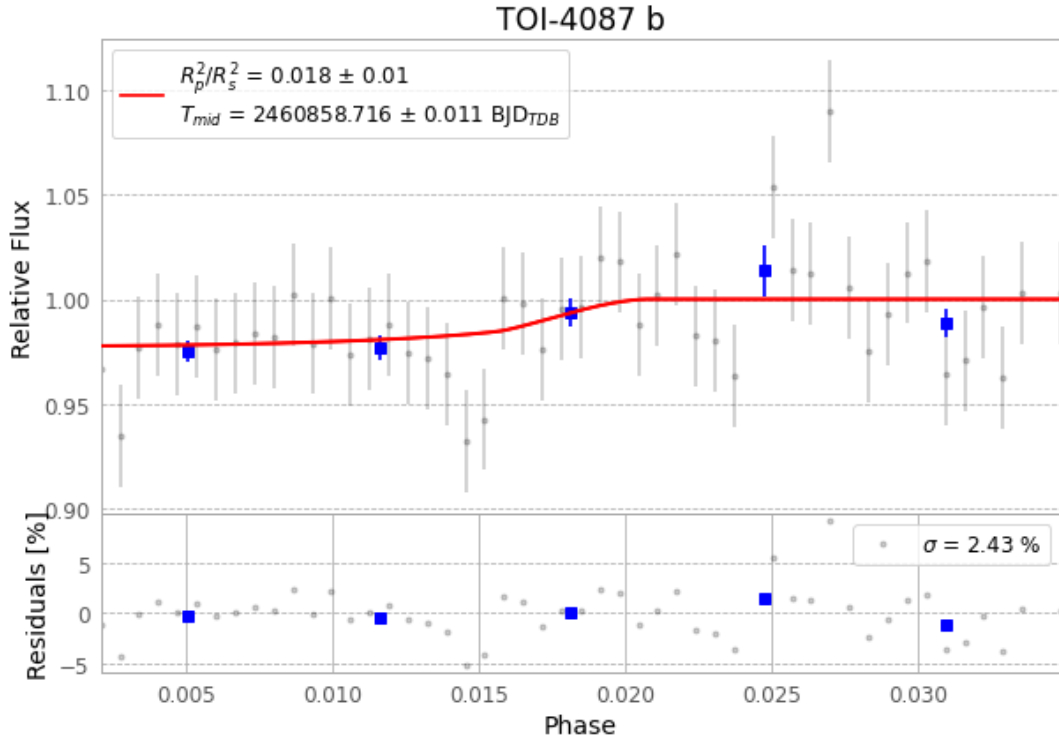


Figure 1 — FinalLightCurve_TOI-4087 b_2025-07-01.png (SHA-256 c7a57ce5f1c840ce...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ed0f3ebb82fddb7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), TOI4087250702052115.FITS → TOI4087250702075416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-250702.log (6522f68af48e...), FinalLightCurve_TOI-4087 b_2025-07-01.png (c7a57ce5f1c8...), FinalLightCurve_TOI-4087 b_2025-07-01.csv (5080cf663d6d...)

Compute resources

Wall time 77.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:58Z.